# A MONOMORPHISM INTO A FIELD-COVERED STACK IS EMPTY OR AN EQUIVALENCE

THE STACKS PROJECT AUTHORS

*Editorial note.* Statement span is lines 2758-2764 of sources/stacks-stacks-properties/pinned-original-source.tex; proof span is lines 2766-2788, the complete author proof (21 body lines).

*Editorial note.* Navigation stubs: 3 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

*Editorial note.* extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

*Editorial note.* Text is retained verbatim, including the author's own typos and placeholders.

*Editorial note.* Queue index 19 of queue_stacks-properties.json (pb=1058, stmt 2758-2764, proof 2766-2788).

*Editorial note.* stacks-properties wave-1: fresh adjudication accept (2026-10-14); pool scanned while CONTROL.md is down (0 bytes) - §2 row deferred per Chief, claim reported to Chief 08:0x.

*Editorial note.* Cross-chapter same-label warning (NOT an occupancy conflict, Chief ruling 2026-10-13 evening): label 'lemma-monomorphism-into-point' also exists in stacks-decent-spaces — a different proposition in a different chapter file.

*Allowed standard prerequisites (editorial).*

- Explicit prerequisite boundary. The cited inputs are prerequisites; the argument in the retained span is not replaced by any cited result.

*Source reference.* Original lemma lemma-monomorphism: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1674–1687. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma lemma-points-cartesian: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 1019–1036. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

*Source reference.* Original lemma 06MG: sources/stacks-stacks-properties/pinned-original-source.tex, source lines 2788–2788. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

**Lemma 0.1.** *Let $\mathcal{Z}' \to \mathcal{Z}$ be a monomorphism of algebraic stacks. Assume there exists a field $k$ and a locally finitely presented, surjective, flat morphism* $\mathrm{Spec}(k) \to \mathcal{Z}$*. Then either $\mathcal{Z}'$ is empty or $\mathcal{Z}' \to \mathcal{Z}$ is an equivalence.*

*Date*: stacks-project:stacks-properties.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

*Proof.* We may assume that $\mathcal{Z}'$ is nonempty. In this case the fibre product $T = \mathcal{Z}' \times_{\mathcal{Z}} \mathrm{Spec}(k)$ is nonempty, see Lemma lemma-points-cartesian. Now $T$ is an algebraic space and the projection $T \to \mathrm{Spec}(k)$ is a monomorphism. Hence $T = \mathrm{Spec}(k)$, see Morphisms of Spaces, Lemma 06MG. We conclude that $\mathrm{Spec}(k) \to \mathcal{Z}$ factors through $\mathcal{Z}'$. Suppose the morphism $z : \mathrm{Spec}(k) \to \mathcal{Z}$ is given by the object $\xi$ over $\mathrm{Spec}(k)$. We have just seen that $\xi$ is isomorphic to an object $\xi'$ of $\mathcal{Z}'$ over $\mathrm{Spec}(k)$. Since $z$ is surjective, flat, and locally of finite presentation we see that every object of $\mathcal{Z}$ over any scheme is fppf locally isomorphic to a pullback of $\xi$, hence also to a pullback of $\xi'$. By descent of objects for stacks in groupoids this implies that $\mathcal{Z}' \to \mathcal{Z}$ is essentially surjective (as well as fully faithful, see Lemma lemma-monomorphism). Hence we win. $\square$